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September 18, 2026

The Editor European Journal of Physics

Dear Editor,

We submit the manuscript “Step size, floating-point precision and the predictability horizon of a simulated double pendulum: a student computational project” for consideration as a Paper in the Mechanics section of *European Journal of Physics*.

The manuscript describes a student computational project, which the journal’s scope names among the contributions it welcomes. It began as our final project in an undergraduate course on numerical analysis. The question is how long a numerical simulation of a chaotic system remains the solution of the equations it was given. By integrating the double pendulum in arbitrary-precision arithmetic we vary the step and the floating-point precision independently, measure the effects of truncation and roundoff separately, and show that the time for which a run is accurate follows a single logarithmic law governed by the largest Lyapunov exponent. The numerical error itself then yields the exponent, in agreement with the standard two-trajectory method.

Intended level. Advanced undergraduate, in physics or applied mathematics. The reader needs the fourth-order Runge–Kutta method, the idea of a floating-point number, and the definition of a Lyapunov exponent.

Usefulness to physics education. The project gives students an operational meaning for the Lyapunov exponent in terms of quantities they control; it separates truncation error from roundoff error, which are usually taught together and rarely distinguished in practice; and it corrects two common beliefs, that a smaller step always gives a better answer and that conservation of energy certifies a trajectory. The program is short, the runs take minutes, and the manuscript sets out how the project can be run with a class and extended.

Relation to previous work in the journal. D’Alessio, *Eur. J. Phys.* **44** (2023) 015002, treats the double pendulum analytically, numerically and experimentally and computes its Lyapunov spectrum. Our manuscript is complementary. It takes the dynamics as given and studies the reliability of the numerical method, which that paper and the earlier pedagogical literature use as a tool.

The manuscript is about 3300 words, with six figures and two tables. Code, data and two animations are supplied as supplementary material. The work is original, has not been published, and is not under consideration elsewhere. Both authors have approved the submission and declare no conflict of interest.

Yours faithfully,

Arjun Pemmasani on behalf of both
authors